

Report - Version 5: Smart Validation

Version 5 extends **Building Agentic AutoML** with a more robust validation strategy.

The agent preserves task detection, feature detection, Data Inspection, the two preprocessing strategies, and the three candidate models from Version 4. The new capability is to evaluate every preprocessing-model combination with **cross-validation** and select the best experiment using the mean score across folds.

Models, metrics, and hyperparameter configurations intentionally remain unchanged so that this version introduces only one new concept: **Smart Validation**.

Goal of Version 5

- preserve the components introduced in Versions 1-4;
- replace the single train-validation split with task-aware validation;
- use **StratifiedKFold** for classification and **KFold** for regression;
- evaluate every preprocessing-model combination across 5 folds;
- store fold scores, mean, and standard deviation for each experiment;
- return `best_preprocessing`, `best_model`, `best_score`, and `best_std`.

Architecture

```
Dataset + Target -> Task/Features -> Data Inspection -> Preprocessing  
Strategies -> Model Candidates -> Smart Validation -> Evaluation -> Best  
Experiment -> State
```

Task Detection

Task detection remains unchanged from Version 4: at most 20 unique target values means classification; otherwise the task is treated as regression.

```
def detect_task(df, target): n_unique = df[target].nunique() if n_unique <= 20: return "classification" return "regression"
```

Feature Detection

Feature detection is also unchanged. On Adult Income, the agent detects 6 numerical features and 8 categorical features.

Data Inspection

The Data Inspector introduced in Version 2 is preserved unchanged. On Adult Income it reports 48,842 rows, 15 columns, two target classes, dtypes, missing values, and cardinality. The main missing values remain workclass = 2,799, occupation = 2,809, and native_country = 857.

Preprocessing

Version 5 preserves the **native** and **impute** strategies. Imputation parameters are learned exclusively from the training fold, preventing leakage between training and validation folds.

Model Selection

LightGBM, XGBoost, and CatBoost remain the three candidates introduced in Version 4. No hyperparameter search is performed, isolating the effect of the new validation strategy.

Metric Selection

Metric selection remains unchanged: ROC AUC for classification and RMSE for regression.

Smart Validation

This is the new capability introduced in Version 5. For classification, the agent selects **StratifiedKFold**; for regression, it selects **KFold**. Both use 5 folds, shuffle=True, and random_state=42. For each experiment, fold_scores, mean_score, and std_score are retained, and final selection uses mean_score.

Training and Evaluation

Every preprocessing-model combination is trained and evaluated across all 5 folds. Preprocessing is refit inside each fold and every fold receives a fresh model instance. This produces a more stable estimate than the single split used in Version 4.

Results - Adult Income

Preprocessing	Model	Mean ROC AUC	Std
native	LightGBM	0.929556	0.002508
native	XGBoost	0.926851	0.002227
native	CatBoost	0.930631	0.002246
impute	LightGBM	0.929217	0.002713
impute	XGBoost	0.926435	0.002643
impute	CatBoost	0.930119	0.002428

Best experiment: **CatBoost + native**, mean ROC AUC = **0.9306306263**, std = **0.0022463105**.

Agent

- load dataset and target;
- detect task and feature types;
- inspect the dataset;
- prepare X and y;
- select the metric and validation strategy;
- build candidate models;
- evaluate every preprocessing x model combination across 5 folds;
- store fold scores, mean, and standard deviation;
- select the best experiment using the mean score;
- return the final state.

```
{ "task": "classification", "metric": "roc_auc", "validation":  
  "StratifiedKFold", "n_splits": 5, "best_preprocessing": "native",  
  "best_model": "CatBoost", "best_score": 0.930630626301291, "best_std":  
  0.002246310473216, "experiments": [...] }
```

Regression Test

The same agent is executed on `simple_regression.csv`. It automatically detects regression, selects KFold, uses RMSE, and evaluates the same three models under both preprocessing strategies.

Preprocessing	Model	Mean RMSE	Std
native	LightGBM	1.843523	0.107316
native	XGBoost	2.015096	0.065982
native	CatBoost	1.663274	0.103853
impute	LightGBM	1.856696	0.120398
impute	XGBoost	2.055328	0.130456
impute	CatBoost	1.628342	0.101940

Best experiment: **CatBoost + impute**, mean RMSE = **1.6283421164**, std = **0.1019398476**.

What Version 5 Teaches

Version 5 introduces the idea that a reliable AutoML choice should not depend on a single split. The agent should estimate performance across multiple folds and retain the variability of the results.

The evolutionary pattern remains explicit: V1 builds the baseline, V2 observes the data, V3 experiments with preprocessing, V4 compares models, and V5 makes validation more robust. The **One Version, One Concept** principle remains intact.

Limitations

- task detection still relies only on the number of unique target values;
- only two preprocessing strategies are available;
- only three boosting model candidates are compared;
- cross-validation is fixed to 5 folds, with no group-aware or time-series-aware logic;
- no hyperparameter optimization;
- no feature engineering, planner, or multi-agent architecture;
- the state stores results and decisions, but not yet a persistent fitted pipeline.

Conclusion

Version 5 replaces the single train-validation split with task-aware cross-validation without changing preprocessing, models, metrics, or hyperparameters. On Adult Income, the best experiment becomes CatBoost + native; on the regression test, CatBoost + impute remains best. The version makes model selection more stable and prepares the project for later AutoML optimization stages.